Extended EDA & Data Pipeline — MILK10k

Computer Vision & Speech Recognition · Homework (Sessions 1–2 follow-up) · Joshua Cole · 08 November 2026

Code: https://github.com/jcole-jpg/Computer-Vision-and-Speech-Recognition — package `src/milk10k/` (`stats`, `color`, `preprocess`, `loader`, `viz`), driver notebook `notebooks/03_eda_pipeline.ipynb`, tests `tests/` (32 pytest cases).

Part 1 — Metadata analysis: correlation with the target

Setup. Target = the 11-class top-level diagnosis (`label` from `training_gt.csv`). The 26 non-id, non-target columns were typed and their missingness measured (table in the notebook). Two are constants, none is free text, and seven are *the label in disguise* (`diagnosis_1..4`, `diagnosis_full`, `melanocytic`, `invasion_thickness_interval`) — excluded from testing and flagged as leakage. Contrary to what the dataset paper suggests, there is **no collection-centre or device field**: `site` is a fine-grained *anatomical* site, and `anatom_site_general` is its coarser copy whose 37 % "missing" rows are exactly the trunk lesions (missing-not-at-random — imputing "unknown" would silently encode *trunk*).

Unit of analysis and statistics. Every lesion has two images with identical patient fields, so lesion-level fields are tested on the 5,240 lesions, not the 10,480 images (which would double every count). Categorical field → Pearson **chi-square** test of independence + bias-corrected **Cramér's V**; numeric / ordinal field → **Kruskal–Wallis H** + **ε²**. Kruskal–Wallis was chosen over one-way ANOVA because age is 5-year-binned and skewed, the MONET scores are bounded in [0, 1] and non-normal, and class sizes run from 9 to 2,522 — the F-test's assumptions fail while the rank test stays valid. With n in the thousands every p-value is ≈ 0, so fields are ranked on effect size; because V is correlation-scale and ε² is variance-scale, √ε² is used for the common ranking.

field	type · level	missing	n	statistic	p	effect size	caveat
`anatom_site_special`	categorical · lesion	98%	103	$\chi^2 = 55$	1e-10	V = 0.70	sparse
`MONET_pigmented`	numeric · image	0%	10,480	H = 4,115	0e+00	$\varepsilon^2 = 0.39$ ($\sqrt{} = 0.63$)	image-derived
`MONET_erythema`	numeric · image	0%	10,480	H = 2,503	0e+00	$\varepsilon^2 = 0.24$ ($\sqrt{} = 0.49$)	image-derived
`age_approx`	numeric · lesion	0%	5,220	H = 1,054	3e-220	$\varepsilon^2 = 0.20$ ($\sqrt{} = 0.45$)	
`MONET_pen_ink`	numeric · image	0%	10,480	H = 1,611	0e+00	$\varepsilon^2 = 0.15$ ($\sqrt{} = 0.39$)	image-derived
`MONET_vasculature_vessels`	numeric · image	0%	10,480	H = 1,203	3e-252	$\varepsilon^2 = 0.11$ ($\sqrt{} = 0.34$)	image-derived
`diagnosis_confirm_type`	categorical · lesion	0%	5,240	$\chi^2 = 514$	4e-104	V = 0.31	bias
`concomitant_biopsy`	boolean · lesion	0%	5,240	$\chi^2 = 514$	4e-104	V = 0.31	bias
`MONET_ulceration_crust`	numeric · image	0%	10,480	H = 1,010	2e-210	$\varepsilon^2 = 0.10$ ($\sqrt{} = 0.31$)	image-derived
`MONET_gel`	numeric · image	0%	10,480	H = 813	4e-168	$\varepsilon^2 = 0.08$ ($\sqrt{} = 0.28$)	image-derived
`skin_tone_class`	ordinal · lesion	0%	5,240	H = 183	6e-34	$\varepsilon^2 = 0.03$ ($\sqrt{} = 0.18$)	bias
`MONET_hair`	numeric · image	0%	10,480	H = 264	6e-51	$\varepsilon^2 = 0.02$ ($\sqrt{} = 0.16$)	image-derived
`site`	categorical · lesion	1%	5,209	$\chi^2 = 744$	1e-118	V = 0.15	
`anatom_site_general`	categorical · lesion	37%	3,284	$\chi^2 = 227$	4e-32	V = 0.14	sparse
`image_manipulation`	categorical · image	0%	10,480	$\chi^2 = 181$	1e-33	V = 0.13	bias
`sex`	categorical · lesion	0%	5,240	$\chi^2 = 73$	1e-11	V = 0.11	

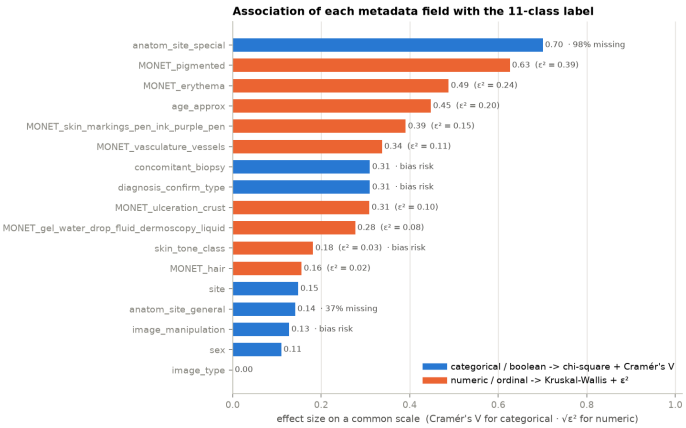


Fig. 1 — Effect size of every usable field, common scale.

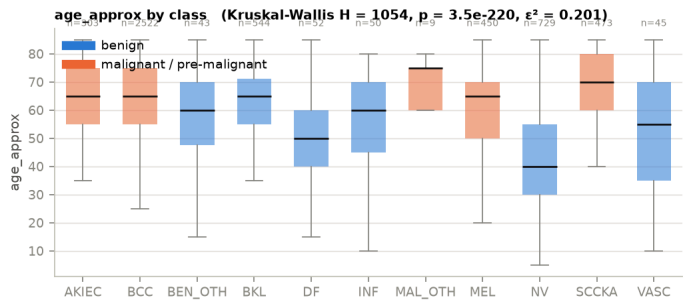


Fig. 2 — Age by class: NV median 40 vs 65–70 for BCC / AKIEC / SCCKA.

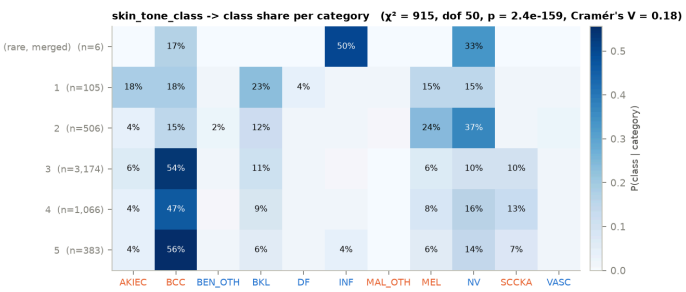


Fig. 3 — P(class | skin tone): tone 2 is 37 % NV / 24 % MEL, tones 3–5 ≈ 50 % BCC.

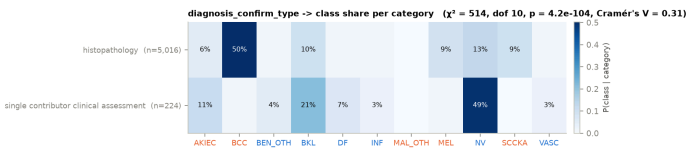


Fig. 4 — Ground-truth method: every non-biopsied lesion is benign — a workflow leak.

Findings. The **most associated usable fields** are (1) **`age_approx`** ($\varepsilon^2 = 0.20$ — median 40 for NV vs 65–70 for the keratinocyte cancers; genuine clinical prior knowledge and a legitimate auxiliary input), (2) the image-derived **`MONET_pigmented`** ($\varepsilon^2 = 0.39$) and **`MONET_erythema`** (0.24) — the strongest signals of all, but they are outputs of a vision-language model run on the image, i.e. image features rather than clinical metadata — and (3) **`skin_tone_class`** / **`site`** ($V \approx 0.15$–0.18: hands are 50 % SCCKA, feet 65 % BCC). **Least useful:** `image_type` ($V = 0$ by construction), `MONET_hair` ($\varepsilon^2 = 0.02$), `sex` ($V = 0.11$: 60 % male overall, 70 % in SCCKA but 50 % in NV) and `anatom_site_special` ($V = 0.70$ looks large but n = 103, 98 % missing — unusable).

> **Bias / leakage flags.** `diagnosis_confirm_type` ≡ `concomitant_biopsy` (identical $\chi^2 = 514$, $V = 0.31$): whether a lesion was biopsied is a decision made *because of* the suspected diagnosis, and every non-biopsied lesion is benign — a model given it learns the clinician's workflow, not the lesion. `image_manipulation` ($V = 0.13$): 327 of 335 "altered" images are clinical photos — an acquisition artefact correlated with class. `skin_tone_class` (only 2 % of lesions are tone 0–1, 61 % tone 3) and `site` are legitimate priors but also proxies for the contributing centre and patient population; their use must be audited with per-tone / per-site metrics.

Part 2 — Colour and histogram analysis across the dataset

Sample. 30 images per class *per modality*, seed 42 (MAL_OTH keeps all 9), 618 images in total; per-image grayscale and R/G/B histograms (64 bins) and mean / std are computed on the raw 0–255 image through the same decode path as the preprocessing pipeline (`milk10k.color`). The two modalities are never pooled: the dermoscopic / clinical gap dominates colour, so a fair class comparison must be made within one modality.

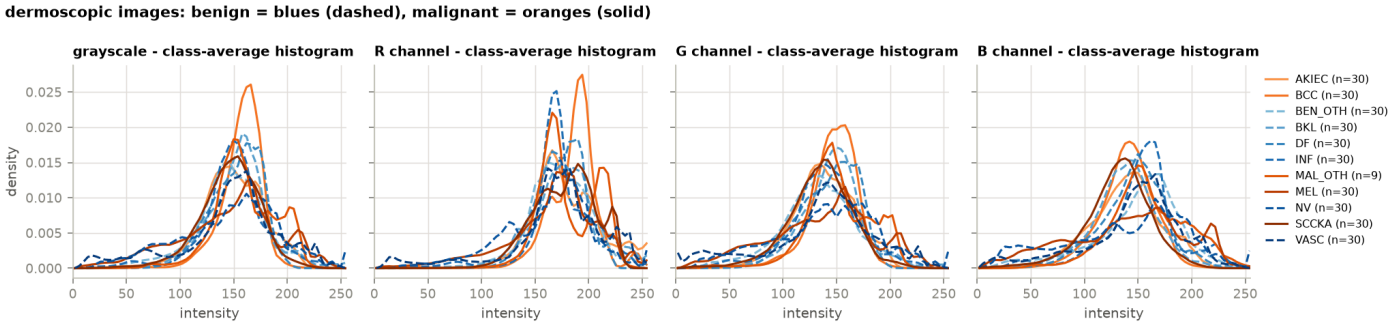


Fig. 5 — Dermoscopic images: class-average grayscale and R / G / B histograms (benign = blues, dashed; malignant = oranges, solid). MEL, NV and VASC carry a broad dark tail; BCC, AKIEC and INF are a single bright peak.

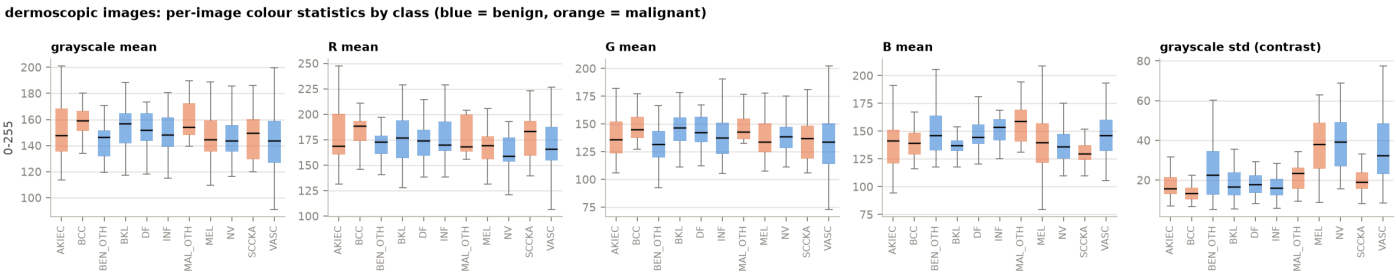


Fig. 6 — Dermoscopic images: per-image colour statistics by class. Channel means overlap; contrast (grayscale std) separates the pigmented classes.

class	n	dermoscopic			clinical		
		gray	std	R / G / B	gray	std	R / G / B
AKIEC	30	150	17	179 / 138 / 138	130	21	175 / 115 / 93
BCC	30	157	14	184 / 146 / 139	149	19	185 / 138 / 111
BEN_OTH	30	146	24	172 / 132 / 148	135	21	164 / 126 / 105
BKL	30	155	19	178 / 146 / 137	140	21	173 / 131 / 105
DF	30	152	18	173 / 143 / 145	141	16	168 / 133 / 110
INF	30	150	16	174 / 138 / 152	145	22	171 / 136 / 122
MAL_OTH	9	161	22	180 / 151 / 159	130	17	162 / 120 / 100
MEL	30	145	37	166 / 136 / 138	142	23	170 / 133 / 113
NV	30	145	39	160 / 139 / 137	146	22	173 / 138 / 115
SCCKA	30	147	20	178 / 134 / 131	131	21	175 / 116 / 92
VASC	30	146	36	168 / 134 / 145	152	23	174 / 145 / 126

Table 2 — Per-class mean of the per-image statistics (0–255).

statistic	ε^2 class (dermoscopic)	ε^2 class (clinical)	ε^2 modality
grayscale mean	0.02	0.07	0.05
grayscale std (contrast)	0.34	0.03	0.00
R mean	0.04	0.02	0.00
G mean	0.04	0.13	0.04
B mean	0.08	0.15	0.32

Table 3 — Share of variance of each per-image statistic explained by class (within one modality) vs by modality (Kruskal–Wallis ε^2).

What differs. Modality dominates: dermoscopic images are bluer and brighter (mean B ≈ 140 vs ≈ 105 in clinical photos; modality alone explains 32 % of B-channel variance) because of the polarised / immersion optics and white-balanced light, whereas clinical photos are warm skin-tone images under room light. **Within dermoscopy, brightness and channel means barely separate classes** (class explains 2–8 % of their variance; every class mean sits between 145 and 161) — but **contrast does**: grayscale std has $\varepsilon^2 = 0.34$, with MEL (37), NV (39) and VASC (36) — pigmented / vascular lesions that put a dark blob on light skin — against BCC (14.5), AKIEC (17) and INF (16), pale lesions with one sharp peak near 160. In clinical photos the pattern flips: contrast is uninformative ($\varepsilon^2 = 0.03$) while G / B means carry 13–15 %: AKIEC / SCCKA photos are darker and redder (gray ≈ 130, B ≈ 93) than VASC / NV / BCC (≈ 146–152, B ≈ 111–126).

Alternative explanations. (i) *Confounding with skin tone and site* (Part 1): BCC / SCCKA / AKIEC come from older, tone-3–5 patients on sun-damaged head / neck / hand skin, whose *background* skin is redder and more textured regardless of the lesion, while NV / MEL cluster in tone 2 on the trunk. (ii) *Acquisition*: dermatoscope model, contact gel, polarisation and white balance differ between the five centres, which also have different case mixes; the dark vignette in some frames lowers the mean with no lesion involvement. (iii) *Framing*: clinical close-ups are not standardised — a 3 mm nevus fills 2 % of the frame, an ulcerated SCC 40 % — so the "image mean" mostly measures how much skin is in frame. (iv) *Sample*: the within-class spread of the means (std 13–28) is as large as the between-class differences, and MAL_OTH has 9 images.

> **Early verdict.** Colour / intensity alone will not classify 11 classes: class means overlap almost completely and MEL vs NV — the clinically critical pair — are indistinguishable on every simple statistic. These features do give a coarse partition, **pigmented / vascular (NV, MEL, VASC — high contrast) vs keratinocytic (BCC, AKIEC, SCCKA, BKL — low contrast)**. The real signal must come from *structure* (pigment network, vessels, borders, texture) and from both modalities together: a learned feature extractor (CNN / ViT) with the modality kept explicit, age as an auxiliary input, and per-modality normalisation so that the modality gap is not the first thing the network learns.

Parts 3–5 — Preprocessing, data loader and visualiser: design

One config, one processing path. `PreprocessConfig(size, color_mode, resize_mode, normalize, mean, std)` is an immutable, validated dataclass; `preprocess_image(src, config) → (array, ImageInfo)` is the only place pixels are decoded, resized and normalised, and the colour analysis, batch function, loader and visualiser all call it, so any change happens once. Resize defaults to an aspect-preserving **centre crop** (MILK10k mixes aspect ratios; stretching warps lesion geometry). Normalisation defaults to **global min–max (÷255 → [0, 1])**, deliberately *not* per-image min–max, which would stretch every image to full contrast and erase exactly the brightness / contrast differences Part 2 measured; `zscore` with dataset-level mean / std is available for training. Every result carries an `ImageInfo` (input / output shape, dtype, value range, steps), so an array is never a mystery.

Failures are data, not exceptions. `preprocess_batch(items, config) → BatchResult(images (N,H,W,C), ids, infos, skipped=[(id, reason)])` catches per-item errors and continues; in the notebook a missing path and a text file are reported as `FileNotFoundError` / `UnidentifiedImageError` while the four good images are stacked.

Loader = table + config + batch size. `available_subset()` drops rows with no file on disk at construction (`loader.n_unavailable`), labels are encoded in `CLASS_CODES` order, and nothing is read until iteration. Because it takes any DataFrame with `isic_id / path / label`, one class serves a dermoscopic-only loader, a train / val split or a debugging subset by filtering the table first. **Visualiser accepts what the loader emits**: `show_image_grid` and `plot_batch_summary` take a `Batch` directly, `plot_class_balance` a DataFrame or a loader, so the sanity check is one line after building a loader. Colour encodes only benign vs malignant; eleven classes are identified by position.

Loader interface

```
cfg = PreprocessConfig(size=(128, 128))
loader = MILK10kLoader(df, cfg, batch_size=32,
        shuffle=True, seed=0, label_col="label")
loader.n_images, loader.n_unavailable  # 10480, 0
loader.classes      # ['AKIEC','BCC',...,'VASC']
len(loader)         # 328 batches per epoch
for batch in loader:          # lazy, on the fly
    batch.images    # (32,128,128,3) float32
    batch.labels    # (32,) int64 -> classes
    batch.label_names, batch.ids  # aligned
    batch.skipped   # [(id, reason)]
loader.class_counts()  # images per class
```

Tests: `python -m pytest tests -q` — 32 cases: shapes / ranges for every config, bad-file skipping, availability filtering, label alignment after a skip, seeded shuffling, statistics on synthetic tables, real MILK10k smoke tests.

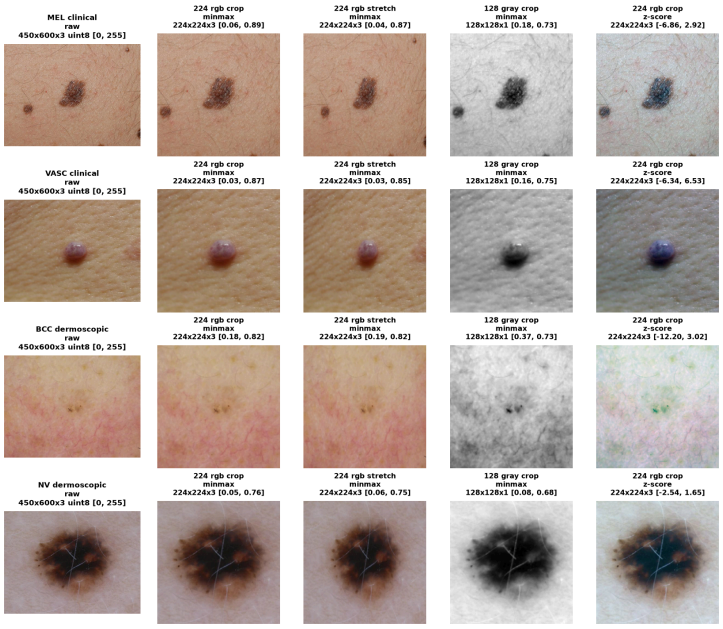


Fig. 7 — Raw vs processed for four real images under four configs: 224 RGB centre-crop min–max, 224 RGB stretch, 128 grayscale, 224 RGB z-score (re-stretched for display).

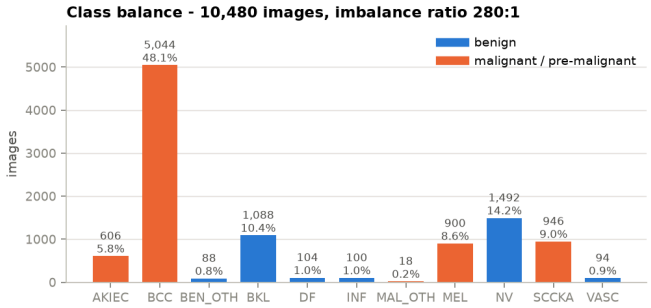


Fig. 8 — Class balance of the available images: BCC 48 %, MAL_OTH 18 images; 280 : 1.

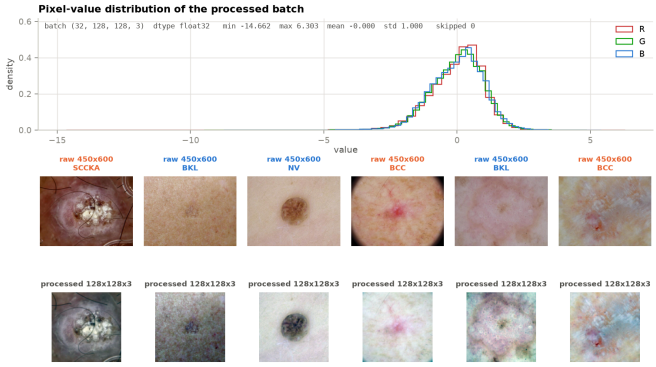


Fig. 9 — Batch summary of a z-score loader batch: mean 0.000, std 1.000, plus raw vs processed.

Class balance. All 10,480 images are available locally. The data is **severely imbalanced**: BCC is 48 % of images, five classes have ≤ 104 images (MAL_OTH 18), a 280 : 1 largest-to-smallest ratio, and 72 % of lesions are malignant. For training this means: split **by lesion** and stratify by class; use class-weighted loss or re-sampling (the loader's `class_counts()` gives the weights) or merge the five tail classes into an "other" bucket first; and evaluate with macro-F1 / balanced accuracy and per-class sensitivity — a model that always says BCC already scores 48 % accuracy.